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<https://doi.org/10.1057/s41599-026-08351-0>

OPEN

Winning the blue sky defense in China: does energy quota trading curb air pollution?

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The energy quota trading (EQT) pilot is a market-based tool for promoting efficient energy allocation. While studies have examined the energy-saving and carbon-reducing benefits of the EQT pilot in China, few have explored its effects on pollution control. As such, our study investigates the effect of EQT on urban air pollution by employing the difference-in-differences (DID) approach and using panel data for 284 Chinese cities from 2013 to 2020. The results indicate that the EQT pilot significantly curbs the pilot city's air pollution. The promotion of urban technological innovation and the optimization of industrial structure are influencing mechanisms, while the mechanism of energy structure optimization is not observed. Furthermore, the pollution reduction effect of EQT varies significantly depending on the geographical location, resource endowment, and financial pressure of cities. The pollution reduction effect of EQT is stronger for central-western cities, energy-oriented cities, and cities with low fiscal pressure. In addition, the EQT pilot has positive spatial spillover effects; i.e., it significantly curbs air pollution in neighboring cities. This study not only expands the research on the environmental impacts of the EQT pilot but also provides empirical support for the design of market-oriented environmental regulatory policies.

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Introduction

Climate warming caused by massive carbon emissions is still intensifying (Chen and Li, 2023) and has become a global problem that seriously threatens the sustainable development of society (Zhao et al. 2023a; Zhao et al. 2026a). China is a major global carbon emitter (Zhao et al. 2021a; Xian et al. 2024), and its total carbon emissions were approximately 11.48 billion tons in 2022, accounting for approximately 27% of global emissions (IEA, 2023). The energy activities behind rapid industrialization and urbanization are important causes of the surge in carbon emissions (Liu et al. 2019). In terms of domestic practices and international experience, energy conservation and energy efficiency are viable means of reducing carbon emissions from energy activities. In 1980, China issued “The Circular on the Gradual Establishment of a Comprehensive Energy Consumption Assessment System”, which gradually established an energy intensity assessment system. During the “Eleventh Five-Year Plan (2006–2010)”, the mandatory assessment of energy consumption intensity was first implemented in China. During the “13th Five-Year Plan (2016–2020)”, the single control of energy consumption intensity was elevated to the dual control of energy consumption intensity and total amount, which put forward stricter requirements on energy utilization. With the tightening of the dual control of energy consumption, several regional governments have used administrative orders to directly intervene in enterprise production, such as limiting production and shutting down enterprises, which leads to an imbalance between supply and demand in the market and increases the uncertainty of enterprises’ expectations. Therefore, stimulating the enthusiasm of enterprises for saving energy through market means and increasing their stable expectations has become a major issue faced by governments. Accordingly, in 2016, China deployed energy quota trading (EQT) pilots in four provinces to promote a more efficient allocation of energy resources and achieve the dual-control targets for energy consumption with the help of market means in energy allocation.

In addition to facing carbon control pressures, China is experiencing severe air pollution (Kuang et al. 2023; Zhou et al. 2020). According to China’s Ministry of Ecology and Environment, in 2022, the average PM_{2.5} and PM₁₀ concentrations in 339 cities at the prefecture level and above were 29 µg/m³ and 51 µg/m³, respectively. These two indicators have declined in recent years but are still well above the WHO recommendations of 5 µg/m³ and 15 µg/m³. Atmospheric pollutants and greenhouse gases are homologous (Jia et al. 2024; Thompson et al. 2014) and mainly originate from fossil fuel energy consumption represented by coal (Wu et al. 2021). Therefore, theoretically, EQT, with the primary goal of energy conservation and carbon control, may also favor air environment improvement. To date, the literature has examined the effects of EQT on energy savings (Wang et al. 2021; Liao et al. 2021), energy vulnerability (Liu and Sheng, 2023), carbon emissions (Du et al. 2023; Chu et al. 2021), and corporate low-carbon technology innovation (Li and Zhao, 2023), but few studies have focused on its air governance effects and spatial spillover effects, leaving a gap for our study to fill. Taken together, the results of this study are valuable in theory and practice as follows. First, this study systematically analyzes the air governance effects of EQT in pilot cities, which expands the research literature on EQT and provides a paradigm for further exploring the environmental impacts of EQT. Second, pollution control and carbon reduction have strategic consistency and tactical coupling and should be managed synergistically. Therefore, systematically exploring the air governance effects of EQT provides a scientific basis for designing synergistic pollution control and carbon reduction governance policies. Third, the Chinese government has underutilized market-based instruments for air pollution

control (Yan et al. 2020) and still adopts mainly administrative instruments (e.g., temporary one- and two-number restrictions and production restrictions for high-polluting firms during special festivals or events). Therefore, studying the effect of the EQT pilot on air pollution provides empirical support for exploring market-based means to combat air pollution. Based on the above, we find that the air governance effect of EQT, which is the focus of this paper, holds significant theoretical and practical value. Whether EQT can curb air pollution is an important question that urgently needs to be answered.

Therefore, based on the above background, this study employs data for 284 Chinese cities from 2013 to 2020 and the difference-in-differences (DID) approach and explores the effect of the EQT pilot on urban air pollution. This work aims to answer the following four questions: First, does the EQT pilot curb the pilot city’s air pollution? Second, if such effects exist, through which path does the EQT pilot affect urban air pollution? Third, does the EQT pilot have an asymmetric effect on pollution abatement depending on the city’s geographic location, resource endowment, and financial pressure? Finally, how does the EQT pilot affect air pollution in surrounding cities? The findings of this paper include the following: First, the EQT pilot effectively curbs pilot cities’ air pollution, reducing both the intensity and the total amount of pollution. Moreover, the pollution-reducing effects of the EQT pilot start to grow as the policy advances. Second, the EQT pilot supports pollution reduction by facilitating urban technological innovation and industrial structure transformation, but the mechanism of energy structure optimization does not exist. Third, the pollution reduction effect of EQT varies significantly depending on the geographical location, resource endowment, and financial pressure of cities. The EQT pilot has a stronger effect on central-western cities, energy-oriented cities, and cities with low fiscal pressure. Finally, the EQT pilot has positive spatial spillover effects; i.e., it significantly curbs air pollution in neighboring cities. It contributes to global pollution reduction without a zero-sum game of environmental governance.

Our work contributes to the available research in the following ways. First, current studies have investigated the effects of EQT on energy consumption (Wang et al. 2021; Zhang and Zhang, 2020; Liao et al. 2021) and carbon emissions (Du et al. 2023; Chu et al. 2021); however, the air governance effect of EQT has thus far garnered little focus. Our work systematically evaluates the effect of EQT on air pollution by employing panel data from 284 Chinese cities, which expands the literature on the policy effectiveness of EQT and enriches the policy options for air pollution governance through market-based means. Second, given the vastness of China’s geography and the large differences among cities, there may be heterogeneity in policy effectiveness. To explore the asymmetric impact of EQT on air pollution, a subsample study is conducted according to geographic location, resource endowment, and financial pressure. These findings provide a basis and reference for the formulation of targeted environmental policies and the improvement of policy efficiency. Third, the literature has focused mostly on the effects of EQT on pilot areas (Yan et al. 2020; Liu and Sheng, 2023) but has not focused on the spatial spillover effects on surrounding cities. Our paper uses the spatial difference-in-differences (SDID) model to explore the effect of the EQT pilot on air pollution in surrounding cities, which provides a scientific basis for more comprehensively understanding the air treatment effect of EQT and coordinating overall pollution treatment.

The remainder of this paper is structured as follows. Section 2 presents the policy background and research hypotheses. Section 3 details the research design. Section 4 reports the empirical

results and analysis. Section 5 provides the heterogeneity and spatial impact analysis. The final section offers a summary.

Policy background and research hypotheses

Policy background. Energy-consuming rights refer to the annual comprehensive energy consumption of energy-consuming units through certification or trading under the premise of total energy consumption and intensity control (Yang et al. 2020). Energy consumption is a kind of special property right, the essence of which is the right of usufruct. EQT is a behavior of enterprises that choose to trade the energy consumption quotas obtained following the law. EQT forces enterprises to improve their energy utilization efficiency by enabling the market to play a decisive role in the allocation of resources. In response to carbon reduction pressures and energy shortages, in July 2016, China released “The Pilot Program on a Compensated Energy-consuming and Trading System”, identifying the provinces of Zhejiang, Fujian, Henan, and Sichuan as the pilot areas for EQT. Under the pilot policy, the right to energy consumption within the quota is mainly free of charge, and the energy consumed over the quota is remunerated. Income from the compensated use of energy-consuming rights should be earmarked for funding inputs in energy conservation, emission reduction, and related work in the region. Pilot regions should improve the price formation mechanism and determine the initial transaction price of energy-consuming rights. Additionally, with market development, the transaction price should be formulated via the aggregated bidding of counterparties. Pilot policies encourage pilot regions to explore the path of EQT according to their characteristics, but they should also incorporate the possibility of connecting with a future nationally unified EQT market into the design of their institutional framework and market mechanism. Both the EQT and the carbon emissions trading pilot schemes implemented in 2011 are aimed at conserving and economizing environmental resources and highlighting the paid use of resources. The EQT and carbon emissions trading pilots are essentially the same, and their implementation effects are highly synergistic, except that the former focuses on front-end management and the latter on end-of-pipe governance (Chu et al. 2021).

In practice, the four pilot provinces explored different paths according to local characteristics. In terms of the path, Zhejiang has adopted the path of “driving the stock with the increment”, with the scope of related transactions covering new energy-consuming projects. Fujian, Henan, and Sichuan have included all incremental and stock enterprises in high energy-consuming industries that have reached a certain threshold of energy consumption in the scope of the pilot program. In terms of the main trading entities, at the initial stage of trading, Zhejiang’s EQT is carried out between energy-consuming enterprises included in the scope of the pilot program and the local government, and is planned to expand to include energy-consuming enterprises when the market matures. In Fujian, Henan, and Sichuan, the participants in the EQT pilot program are mainly high-energy-consuming enterprises. In addition, in terms of indicator allocation, Zhejiang stipulates that any enterprise with new energy consumption per unit of industrial added value higher than 0.6 t of standard coal per 10,000 yuan should purchase energy-consuming indicators from the government. Fujian, Henan, and Sichuan have adopted a combination of initial nonreimbursable allocation and reimbursable trading based on carbon emissions trading.

Hypotheses

EQT and urban air pollution. EQT, as a kind of market-based environmental regulation, solves the externality problem of

energy consumption through the paid use of scarce environmental capacity to realize Pareto-optimal energy resource allocation. The EQT constrains enterprise production at the source of energy, forcing enterprises to control energy consumption, improve energy efficiency, and use renewable energy. Although EQT is aimed mainly at saving energy and controlling carbon emissions, it may also have a synergistic effect on air pollution. Energy inefficiency is an important cause of air pollution in China (Liu et al. 2018; Liu et al. 2020). The link between EQT and green development is formed through energy efficiency. EQT can facilitate the flow and agglomeration of energy to high-quality projects, enterprises, industries, and developed regions to promote improvements in energy efficiency, thus reducing air pollution. On the other hand, carbon emissions exacerbate climate warming and reduce the difference in vertical temperature between the boundary layers of the atmosphere, resulting in the formation of an inversion layer. The inversion layer makes it more difficult for atmospheric pollutants to spread, thus aggravating air pollution. Moreover, carbon dioxide can react with nitrogen oxides and volatile organic compounds in the atmosphere to form ozone and particulate matter, thus aggravating air pollution. EQT is expected to curb air pollution by curbing carbon emissions to increase the dispersion of pollution and reduce associated chemical reactions. Therefore, unlike carbon emissions trading and pollution rights trading, which suppress pollution from the end of the process (Dong et al. 2022; Shen et al. 2023), EQT can suppress air pollution at the source. Thus, we develop hypothesis 1.

Hypothesis 1: The EQT pilot curbs air pollution in pilot cities.

EQT, technological innovation, and urban air pollution. Technological innovation is important for controlling air pollution (Zhang et al. 2017). EQT is expected to curb urban air pollution by promoting technological innovation. Specifically, environmental regulation exerts environmental pressure on firms, forcing them to change their strategies (Yu et al. 2023). The Porter hypothesis proposes that appropriate environmental regulation stimulates firms to increase technological investment (Porter and Linde, 1995). As a market-based environmental regulation, EQT increases the cost of energy consumption for firms with low energy efficiency in the pilot region. These enterprises often need to purchase energy-consuming rights to stabilize production. The increased demand for energy rights will increase their prices, which in turn increases the cost of energy for enterprises and affects the competitiveness of their products. Consequently, this forces enterprises to strengthen technological innovation to save energy and reduce consumption. Enterprises with a high level of energy efficiency can obtain additional income by selling residual energy-consuming rights, which is conducive to incentivizing enterprises to enhance technological transformation. As a result, this creates a positive cycle. In addition, the EQT pilot proposes increasing financial and taxation efforts to support energy-saving reforms in enterprises. Fiscal policy follows the government’s policy orientation and strategic intent and can alleviate the lack of funds for the energy-saving research and development (R&D) of enterprises and further stimulate enthusiasm for innovation (Howell, 2017). Accordingly, our study develops hypothesis 2:

Hypothesis 2: The EQT pilot suppresses pilot cities’ air pollution by facilitating technological innovation.

EQT, industrial structure upgrading, and urban air pollution. Industrial restructuring is an important means of optimizing resource allocation and controlling pollution (Chen and Zhao, 2019), and EQT is expected to curb pilot cities’ air pollution by facilitating industrial structure upgrading. More specifically, although China’s new industries have recently boomed, the

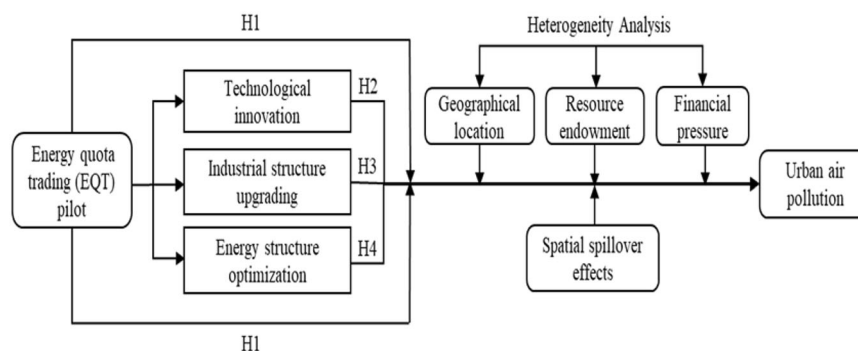


Fig. 1 Logical framework.

industrial structure, which has a high proportion of heavy industry, has not fundamentally changed. The energy structure is dominated by coal (Luo et al. 2019), and an industrial structure with a high proportion of heavy chemical industries entails the consumption of enormous amounts of energy, leading to continuous pressure on urban air pollution. The EQT pilot tightens the production constraints of firms in pilot cities, driving enterprises with backward production capacity that cannot be transformed to downsize, relocate, or even exit the market. The downsizing of backward production capacity will alleviate the biased industrial structure of the city, thus improving the urban air environment. Moreover, the resources released from energy-intensive industries may be transferred to tertiary industries, which are generally considered more low-carbon and environmentally friendly (Wang et al. 2019). The industrial structure of pilot cities is thus gradually optimized through resource reallocation. In terms of the practices of the pilot regions, all four pilot provinces focused on EQT in high energy-consuming industries. Heavy industries with high energy consumption are the main cause of air pollution (Cheng and Kong, 2022); thus, EQT can limit the broad expansion of energy-consuming industries and force them to accelerate green transformation. Summarizing the above analysis, our paper develops hypothesis 3:

Hypothesis 3: The EQT pilot curbs urban air pollution by facilitating industrial structure upgrading.

EQT, energy structure optimization, and urban air pollution. Upgrading energy cleanliness and optimizing the energy structure are important directions for curbing air pollution. EQT is expected to curb urban air pollution by improving energy cleanliness and optimizing the energy structure. Specifically, China's resource endowment, which is rich in coal, poor in oil, and low in gas, has led its energy consumption to be dominated by coal, which is another main contributor to air pollution (Dong et al. 2019; Wang et al. 2019). Although the pattern dominated by coal does not change in the short term, China must gradually reduce its proportion of coal consumption to further control air pollution (Shen et al. 2019). Therefore, while making good use of coal cleanly and efficiently, China should gradually increase the proportion of renewable energy consumption in its energy mix. It has also been confirmed that nonrenewable energy increases air pollution and that renewable energy helps reduce air pollution (Chen et al. 2023a). The EQT pilot stipulates that an enterprise's self-produced and self-consumed renewable energy is not included in its overall energy consumption. Thus, to reduce the cost of energy consumption or gain revenue by selling more surplus energy quotas, under the guidance of the EQT policy, enterprises have the incentive to develop and use more renewable energy. Renewable energy sources include wind, solar, hydro, and biomass (Doğan et al. 2020), which are clean and less harmful to the

environment (Hu et al. 2023a). In addition, firms' investment in renewable energy sources has positive effects on the energy mix (Doğan et al. 2020), which enhances the efficiency of fossil fuel utilization and further reduces energy consumption (Miremedi et al. 2019). Accordingly, this paper proposes hypothesis 4:

Hypothesis 4: The EQT pilot suppresses urban air pollution by promoting energy structure optimization.

The logical framework of the study is presented in Fig. 1.

Methods and data

Empirical model. This study regards China's EQT pilot in 2017 as a quasinatural experiment and aims to evaluate its effect on urban air pollution. Following previous literature (Zhao et al. 2022; Chen et al. 2023b; Guo et al. 2024a), this study uses the DID model to shed light on the net effect of the EQT pilot. The benchmark regression model is shown in Eq. (1):

$$Pollution_{it} = \alpha_0 + \alpha_1 Treat_i \times Post_t + \gamma Control_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (1)$$

where the subscripts i and t denote the city and year, respectively. The dependent variable ($Pollution$) is the air pollution of city i in year t . The core independent variable is the interaction term $Treat \times Post$, and its regression coefficient α_1 is the focus of the regressions, as it represents the effect of the EQT policies on urban air pollution. If $\alpha_1 < 0$, then the EQT pilot suppresses urban air pollution; otherwise, it does not. $Control$ denotes multiple control variables. μ and τ represent city and year fixed effects, respectively. α_0 and ε are constant and random error terms, respectively.

Variable selection

Dependent variable. Air pollution ($Pollution$) is the dependent variable. We evaluated urban air pollution in terms of two dimensions: industrial soot emission intensity ($Soot1$, the logarithm of the ratio of industrial soot emission to industrial value added) and industrial soot emission ($Soot2$, the logarithm of total industrial soot emission). There are two main reasons for choosing industrial soot to characterize air pollution. First, industrial soot, which is fine particulate matter produced during fuel combustion, is an important cause of hazy weather, so controlling industrial soot is crucial to urban air improvement. Second, coal is the dominant primary energy source in China (Wang et al. 2019), and its main pollutants are sulfur dioxide, carbon dioxide, and soot. To prevent interference from the sulfur dioxide emissions trading policy and carbon emissions trading policy in the regression, it is reasonable to choose industrial soot to measure air pollutant emissions (Song et al. 2022). Additionally, in a robustness test, we choose industrial sulfur dioxide (emission intensity and total emissions) and the urban $PM_{2.5}$ index to further verify the reliability of the core findings.

Table 1 Summary statistics.

Variables set	Variables	Obs	Mean	S.D.	Min	Max
Dependent variables	<i>Soot1</i>	2272	3.112	1.352	−0.710	6.070
	<i>Soot2</i>	2272	9.534	1.184	6.269	12.183
Independent variable	<i>Treat × Post</i>	2272	0.096	0.295	0.000	1.000
Control variables	<i>Industry</i>	2272	0.442	0.104	0.163	0.692
	<i>PGDP</i>	2272	1.577	0.527	0.411	2.803
	<i>Population</i>	2272	1.277	0.761	−0.901	2.790
	<i>Urb</i>	2272	0.568	0.143	0.282	0.950
	<i>Open</i>	2272	0.015	0.015	0.000	0.070
	<i>Finance</i>	2272	2.561	1.129	1.096	6.801

Independent variable. The interaction term *Treat × Post* is the independent variable. *Treat* and *Post* represent the city grouping and time grouping variables, respectively. *Treat* is set to 1 when the city is a pilot city; otherwise, it is set to 0. *Post* is set to 1 when the year is 2017 or later; otherwise, it is set to 0. In September 2016, China launched an EQT pilot in Zhejiang, Fujian, Henan, and Sichuan Provinces. Therefore, we consider the 55 prefecture-level cities under the jurisdiction of the four provinces as pilot cities. Considering the time lag of policy impacts, our study sets 2017 as the starting year of policy implementation.

Control variables. In accordance with earlier studies (Hu et al. 2021; Dong et al. 2022; Ai et al. 2022; Wang et al. 2023b; Wu et al. 2023; You et al. 2024), we incorporate the following variables as control variables into the baseline regression: urban industrial structure (*Industry*, the proportion of the added value of the secondary industry in GDP), urban economic development (*PGDP*, the logarithm of the per capita GDP), population density (*Population*, the logarithm of the ratio of the urban population to urban area), urbanization (*Urb*, the ratio of the urban population to total population), openness (*Open*, the ratio of actual utilization of foreign capital to GDP), and financial development (*Finance*, the ratio of total deposits and loans of commercial banks to GDP). The selection of these control variables is primarily based on the following reasons: The secondary industry is a major source of energy consumption and pollutant emissions. If a city's economic structure is predominantly industrial, its air pollution levels may be inherently higher. Controlling for this variable helps isolate the policy effect from the inherent differences in pollution levels caused by variations in industrial structure. Economically developed regions may possess more abundant resources for environmental governance. Therefore, controlling for the level of economic development is crucial to avoid omitted variable bias in the model. A higher population density typically implies more concentrated production, consumption activities, and traffic flow, which can lead to more intensive pollutant emissions. Controlling for this variable helps capture the impact of pollution resulting from different degrees of population agglomeration. The urbanization process is accompanied by significant changes in energy consumption patterns and land use, which are often associated with increased energy demand and emissions. This variable reflects the systematic impact of urban development stages and social structure on environmental pollution. Foreign investment may exacerbate environmental pressure in host countries through the “pollution haven” effect, or it may generate a “pollution halo” effect by introducing advanced technologies. Controlling for this variable is intended to capture the potential impact of international capital flows on the urban environment. A developed financial system can provide funding for corporate green innovation and pollution

control, potentially contributing positively to environmental quality improvement. Neglecting this factor may lead to misjudgment of the policy's actual effect.

Data specification. This paper takes 2013–2020 as the study interval and 284 Chinese cities (55 pilot cities and 229 nonpilot cities) as the object. The study period ends in 2020 because, after 2021, the remaining nonpilot provinces actively responded to the requirements of the *National 14th Five-Year Plan* to “promote the market-oriented trading of energy-consuming rights” and carry out the corresponding practices. Therefore, we consider the years until 2020 to prevent this change from affecting our results. The data used in this paper are from the China City Statistical Yearbook and the China Industrial Statistical Yearbook. We deflate all the nominal indicators based on price indices. Table 1 displays the summary statistics of the main variables. With respect to the dependent variables, the mean value of *Dust1* is 3.112, yet it has a substantial standard deviation (1.352), and the range between the minimum (−0.710) and maximum (6.070) values is markedly wide. This finding indicates significant heterogeneity in industrial soot and dust emission intensity among Chinese cities, with some cities exhibiting extremely high pollution emission intensity and others demonstrating effective control. For *Dust2*, the mean value is 9.534, and its distribution is relatively concentrated (S.D. = 1.184). However, the extreme values reveal that the city with the highest emission volume is approximately 370 times that of the city with the lowest, highlighting a pronounced disparity in the total pollution load among cities. The above analysis preliminarily suggests that the sample in this study encompasses cities with diverse development patterns and pollution characteristics, thereby providing a research foundation with sufficient variability for subsequently identifying the net effect of the EQT policy.

Empirical results and analyses

Preliminary comparative analysis before and after the pilot.

Table 2 reports a comparison of key variables for pilot and nonpilot cities. For *Soot1* and *Soot2*, the mean test shows that the pilot cities have significantly smaller values than the nonpilot cities in both the nonpilot (2013–2016) and pilot (2017–2020) periods, with the difference increasing from 0.651 to 1.028 and 0.302 to 0.572, respectively. Furthermore, to visually compare the changes in air pollution before and after the EQT pilot, Figs. 2, 3 show the changes in *Soot1* and *Soot2*; the pollution levels in the pilot and nonpilot cities have almost identical trends before the EQT pilot, whereas the pollution gap increases significantly after the pilot shock. Therefore, without controlling for other factors, it can be tentatively concluded that the EQT pilot is likely to have suppressed pilot cities' air pollution. However, this conclusion still requires rigorous testing.

Parallel trend assumption test. Satisfying the parallel trend assumption is a prerequisite for employing the DID approach for policy assessment. Although the trends of the two main dependent variables (*Soot1* and *Soot2*) before 2017 are essentially parallel (see Figs. 2, 3), a more rigorous test is still needed. Thus, drawing on Zhao et al. (2023), we build the following event study model (see Eq. (2)) with the 4th period (i.e., 2013) before 2017 as the base period. To prevent collinearity, the base period is eliminated from the test.

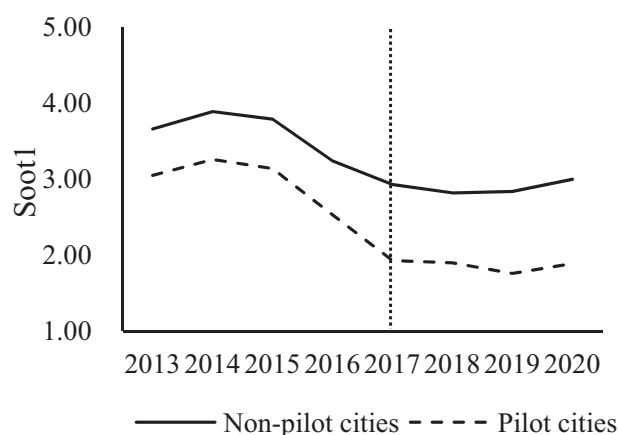
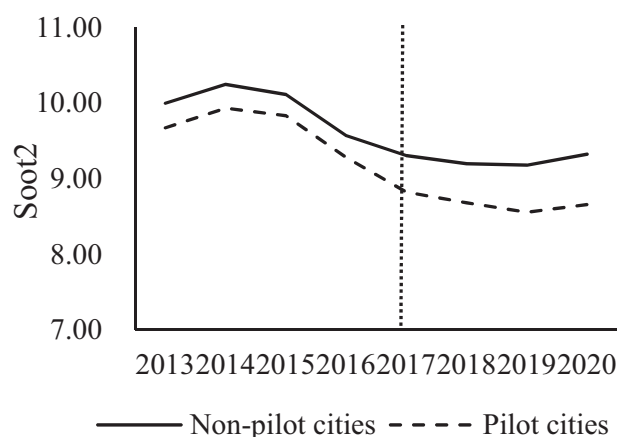
$$Pollution_{it} = \beta_0 + \sum_{j=-3}^3 \varphi_j Treat_i \times Year_{t+j} + \gamma Control_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (2)$$

where $Year_{t+j}$ denotes the dummy variable for the j th period of the policy shock, including the 3 periods before and 3 periods

Table 2 Comparison of key variables before and after policy shocks.

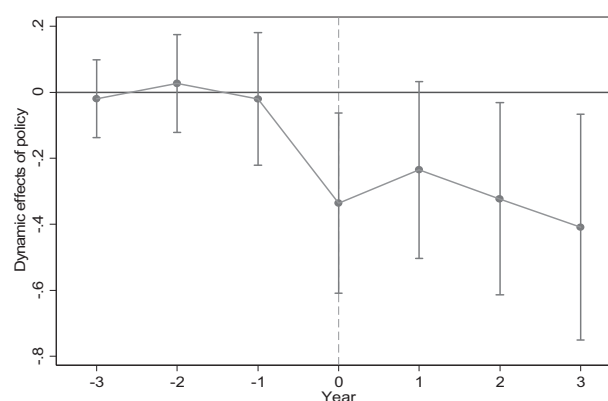
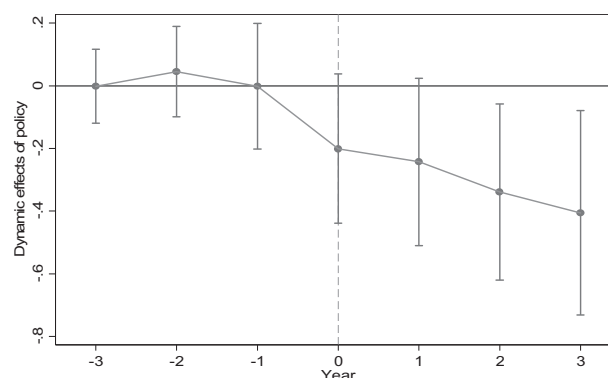
Variables	Nonpilot period (2013–2016)			Pilot period (2017–2020)		
	Nonpilot cities	Pilot cities	Mean test	Nonpilot cities	Pilot cities	Mean test
<i>Soot1</i>	3.644	2.992	0.651***	2.898	1.870	1.028***
<i>Soot2</i>	9.984	9.682	0.302***	9.252	8.680	0.572***
<i>Industry</i>	0.466	0.519	−0.053***	0.401	0.447	−0.047***
<i>PGDP</i>	1.467	1.485	−0.017	1.656	1.790	−0.134***
<i>Population</i>	1.286	1.580	−0.294***	1.168	1.383	−0.215***
<i>Urb</i>	0.549	0.519	0.031***	0.599	0.572	0.027***
<i>Open</i>	0.017	0.017	0.000	0.014	0.017	−0.003*
<i>Finance</i>	2.369	2.285	0.084	2.858	2.435	0.423***

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**Fig. 2** Comparison of pollution levels in pilot and nonpilot cities (*Soot1*).**Fig. 3** Comparison of pollution levels in pilot and nonpilot cities (*Soot2*).

after the policy shock. The significance of ϕ_j represents whether a parallel trend exists between the pilot and nonpilot areas. According to the results in Figs. 4, 5, the coefficients of ϕ are not significant for the period 2014–2016, revealing that the pilot cities have similar operational trends as the nonpilot cities do, which satisfies the parallel trend assumption. In addition, the data in Figs. 4, 5 suggest that with the advancement of EQT policies, the inhibitory effect of EQT on air pollution gradually strengthens.

Effects of the EQT pilot on urban air pollution. In accordance with Eq. (1), Table 3 details the benchmark estimation results of the effect of the EQT pilot on urban air pollution. Specifically,

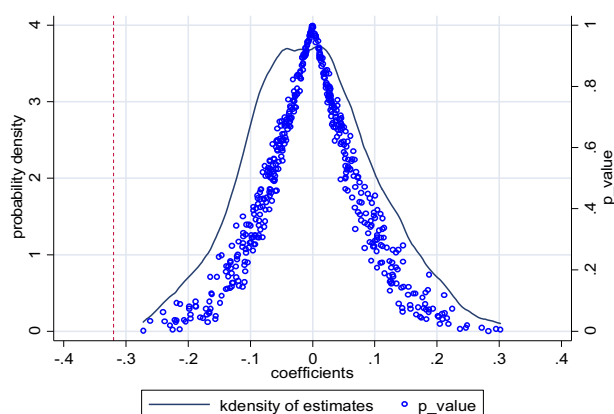
**Fig. 4** Parallel trend test (*Soot1*).**Fig. 5** Parallel trend test (*Soot2*).

columns (1) and (3) contain only the core independent variable; the regression coefficients of $Treat \times Post$ on pollution intensity and total pollution are significantly negative ($\alpha_1 = -0.385$, $p < 0.01$; $\alpha_1 = -0.281$, $p < 0.01$). Columns (2) and (4) detail the results of including control variables; the coefficients of $Treat \times Post$ are both still significantly negative ($\alpha_1 = -0.320$, $p < 0.01$; $\alpha_1 = -0.304$, $p < 0.01$). Therefore, the estimated baseline results indicate that the EQT pilot effectively curbs air pollution in pilot cities, confirming Hypothesis 1. This core finding is also similar to that of Qi and Han (2021), who concluded that the EQT pilot can reduce emissions in terms of total amount and intensity. From an economic perspective, taking the most rigorously specified models in columns (2) and (4) as examples, the EQT pilot led to a reduction in the pilot cities' pollution intensity and total pollution by approximately 27.38% (i.e., $e^{-0.320} - 1$) and 26.21% (i.e., $e^{-0.304} - 1$), respectively. In terms of the control variables, the

Table 3 Benchmark estimation of the effect of the EQT pilot on urban air pollution.

Variables	Soot1 (1)	Soot1 (2)	Soot2 (3)	Soot2 (4)
<i>Treat × Post</i>	−0.385*** (−3.38)	−0.320*** (−2.96)	−0.281*** (−2.73)	−0.304*** (−2.96)
<i>Industry</i>		−2.915*** (−4.80)		−0.309 (−0.55)
<i>PGDP</i>		−0.916*** (−4.94)		−0.212 (−1.17)
<i>Population</i>		0.084 (0.65)		0.156 (1.25)
<i>Urb</i>		0.084 (0.16)		0.153 (0.30)
<i>Open</i>		−0.368 (−0.16)		−1.292 (−0.62)
<i>Finance</i>		−0.097* (−1.88)		−0.182*** (−3.40)
<i>Cons</i>	3.150*** (285.44)	5.981*** (12.40)	9.561*** (957.55)	10.236*** (20.88)
<i>Obs</i>	2272	2272	2272	2272
<i>R²</i>	0.843	0.858	0.824	0.826
<i>City FE</i>	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y

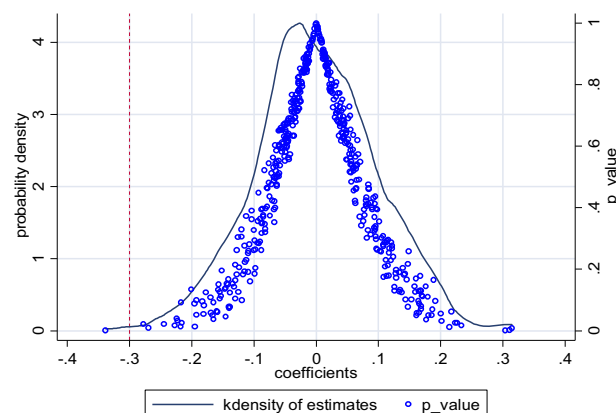
Note: (1) *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; (2) The values in parentheses are the t-statistics corresponding to cluster-robust standard errors at the city level. Unless otherwise specified, this note applies to the following tables.

**Fig. 6** Urban placebo test (Soot1).

coefficients of *Finance* in columns (2) and (4) are significantly negative, indicating that finance has effectively suppressed urban pollution, which is consistent with expectations. The coefficients of *Population*, *Urb*, and *Open* are not significant at conventional levels, suggesting that these three factors do not have a substantial effect on urban pollution. Both urban industrial structure and economic development have contributed to reducing pollution intensity, but they have not shown a significant inhibitory effect on total pollution; this may be because the expansion of economic scale has offset the emission reduction effects brought about by the decline in pollution intensity.

Robustness checks

Placebo test. A possible challenge to the benchmark regression is that the statistical significance of *Soot1* and *Soot2* may result from unknown factors. Thus, to examine whether the air governance effects of EQT policies are influenced by random factors, in reference to prior papers (Zhao et al. 2023b; Guo et al. 2024b; Zhao et al. 2024a; Zhao et al. 2024b; Chen et al. 2025), this study

**Fig. 7** Urban placebo test (Soot2).

performs a city placebo test by randomly generating treatment groups. Specifically, consistent with the number of real pilot cities (55), the policy effect is reassessed by randomly selecting 55 cities from the full sample as dummy pilot cities and the remaining 229 cities as dummy nonpilot cities. This extraction process is repeated 500 times. The kernel density distributions of the regression coefficients (α_1) and the corresponding p values obtained from the placebo test are shown in Figs. 6, 7. Most of the regression coefficients are clustered around the value of 0, and the regression coefficients in the baseline estimation (−0.320 and −0.304) are outliers and small probability events in the urban placebo test. Accordingly, it can be ruled out that the benchmark regression results are affected by unknown factors.

Propensity score matching (PSM)-DID. In the PSM-DID method, units are first matched by PSM to obtain a comparable and stable control group, and then DID regression is used to calculate the causal effect of the policy shocks, thus solving the selection bias brought about by observable and unobservable variables and ensuring the validity of the DID approach. Specifically, we use urban industrial structure, economic development, population density, urbanization, openness, and financial development as covariates to match the cities in the treatment and control groups; then, the matched results are subjected to a DID regression. In columns (1) and (2) of Table 4, the results of the PSM-DID are presented; the regression coefficients for *Treat × Post* are still significantly negative after the sample bias issue is addressed, and the core conclusions remain unchanged.

Removing special samples and considering implementing lag one-stage. First, municipalities and provincial capitals have higher administrative levels, environmental requirements, and economic development, which may affect the reliability of the baseline estimation. Thus, we exclude the 30 municipalities and provincial capital cities from the sample. As reflected in columns (3)–(4) in Table 4, the estimated coefficients change very little compared with the baseline estimation, revealing that the special sample does not have a destructive effect on the benchmark results. Third, following Zhao et al. (2026b), we lagged both the core explanatory variables and the control variables by one period to mitigate the impact of endogeneity on the regression results. Columns (5) and (6) in Table 4 present the corresponding results; the core conclusions remain valid.

Replacing the dependent variable. Different pollution indicators may lead to different findings (Han et al. 2021). As such, we further validate the air governance effect of the EQT pilot by replacing the dependent variables of the benchmark regression

Table 4 Results of PSM-DID, removing special samples, and considering implementing lag one-stage.

Variables	Soot1 (1)	Soot2 (2)	Soot1 (3)	Soot2 (4)	Soot1 (5)	Soot2 (6)
<i>Treat × Post</i>	−0.422*** (−2.97)	−0.394*** (−2.87)	−0.321*** (−2.82)	−0.304*** (−2.81)	−0.225** (−2.46)	−0.277*** (−3.00)
Cons	5.775*** (11.88)	10.052*** (20.27)	6.245*** (12.55)	10.374*** (20.16)	5.627*** (10.09)	10.183*** (18.62)
Obs	2009	2009	2032	2032	1988	1988
R ²	0.863	0.832	0.855	0.822	0.888	0.865
Control	Y	Y	Y	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

Table 5 Regression results for replacing the dependent variable.

Variables	Sulfur1 (1)	Sulfur1 (2)	Sulfur2 (3)	Sulfur2 (4)	PM _{2.5} (5)	PM _{2.5} (6)
<i>Treat × Post</i>	−0.338*** (−3.59)	−0.272*** (−3.07)	−0.227*** (−2.66)	−0.253*** (−2.96)	−0.032*** (−3.12)	−0.023** (−2.25)
Cons	3.275*** (358.56)	5.252*** (11.10)	9.691*** (1170.19)	9.546*** (21.16)	3.646*** (3642.65)	3.636*** (46.94)
Obs	2272	2272	2272	2272	2272	2272
R ²	0.891	0.890	0.894	0.878	0.960	0.955
Control	N	Y	N	Y	N	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

Table 6 Excluding urban environmental governance, other environmental policies, and considering the Kuznets curve.

Variables	Soot1 (1)	Soot2 (2)	Soot1 (3)	Soot2 (4)	Soot1 (5)	Soot2 (6)	Soot1 (7)	Soot2 (8)
<i>Treat × Post</i>	−0.327*** (−3.04)	−0.308*** (−3.00)	−0.324*** (−3.00)	−0.306*** (−2.98)	−0.318*** (−2.94)	−0.300*** (−2.93)	−0.322*** (−2.93)	−0.309*** (−2.95)
Cons	5.744*** (11.79)	10.076*** (20.17)	5.904*** (12.25)	10.183*** (20.73)	5.911*** (12.07)	10.129*** (20.57)	6.010*** (11.40)	10.322*** (19.21)
Obs	2272	2272	2272	2272	2272	2272	2272	2272
R ²	0.876	0.849	0.876	0.849	0.858	0.827	0.858	0.826
EG1	Y	Y	N	N	N	N	N	N
<i>EG1 × Post</i>	Y	Y	N	N	N	N	N	N
EG2	N	N	Y	Y	N	N	N	N
<i>EG2 × Post</i>	N	N	Y	Y	N	N	N	N
LCCP	N	N	N	N	Y	Y	N	N
ESER	N	N	N	N	Y	Y	N	N
Control	Y	Y	Y	Y	Y	Y	Y	Y
City FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y

with industrial sulfur dioxide emission intensity (*Sulfur1*), industrial sulfur dioxide emissions (*Sulfur2*), and the annual average PM_{2.5} concentration (*PM_{2.5}*) in cities. As summarized in Columns (1)–(6) of Table 5, after the dependent variables are replaced, the EQT policy is still effective at suppressing air pollutants, and the reliability of the core conclusion is further enhanced.

Removing the impact of urban environmental governance. Urban environmental governance is a significant factor influencing pollution emissions. Therefore, we incorporate urban environmental governance into the regression to isolate its corresponding impact. Specifically, we counted the frequency of environment-related keywords (i.e., environmental protection, pollution, energy consumption, emission reduction, sewage discharge,

ecology, green, low-carbon, air, chemical oxygen demand, sulfur dioxide, carbon dioxide, PM₁₀, and PM_{2.5}) in the government work reports of each city, added 1, and took the logarithm, denoted as *EG1*. Columns (1) and (2) in Table 6 present the regression results after *EG1* and *EG1 × Post* are added to the baseline model. Furthermore, to eliminate the influence of the text length of government work reports on the measurement of urban environmental governance, we recalculated environmental governance using the proportion of environment-related keywords to the total word count of the reports, denoted as *EG2*. Columns (3) and (4) in Table 6 present the regression results after adding *EG2* and *EG2 × Post* to the baseline model. Compared with the baseline regression, the coefficients of *Treat × Post* in columns (1)–(4) of Table 6 show minimal changes. Thus, the core conclusions remain valid.

Removing other environmental policies and considering the impact of the Kuznets curve. First, other environmental policies may overlap with the EQT policy and thus have an impact on the benchmark regression results. By searching for environmental policies during 2013–2020, we found that two policies may affect the stability of the baseline regression, namely, the three batches of low-carbon city pilots (LCCP) released in 2010–2017 and the three batches of comprehensive demonstration cities for energy conservation and emission reduction fiscal policies (ESER) implemented in 2011–2014. When a city is selected for these two types of environmental pilots, LCCP or ESER is set to 1; otherwise, it is set to 0. We include dummy variables for both types of policies in the baseline estimates to control for potential effects. As shown in columns (5)–(6) of Table 6, the coefficients of $Treat \times Post$ change little compared with those in the benchmark regression, suggesting that these environmental policies do not affect the benchmark results. Second, the environmental Kuznets curve points to an inverted U-shaped phenomenon in which regional environmental pollution may worsen as per capita income increases. As such, drawing on Zhao et al. (2022), we add the square of the PGDP to Eq. (1) to control for the effect of the environmental Kuznets curve on the benchmark regression. As displayed in columns (7)–(8) of Table 6, the core findings remain unchanged.

Sensitivity testing of the baseline model. The EQT pilot policy was not implemented in individual cities but was decided upon and rolled out at the provincial level; this means that all cities within the same province and the same year were subjected to the same policy shock. As a result, the error terms of these cities are likely correlated. Such correlation would underestimate the true standard errors, leading to an overestimation of the t-statistics. Clustering at the “province $\times$ year” level treats each province in each year as an independent cluster. This approach allows for

correlated error terms only among cities within the same province and the same year, while assuming independence across different years for the same province. This is theoretically more reasonable and rigorous. Table 7 presents the corresponding regression results; the coefficients of $Treat \times Post$ remain significantly negative at the 1% level. Therefore, regardless of the clustering assumption, the policy effect persists.

Mechanism identification. The benchmark regression and robustness checks suggest that the EQT pilot can significantly suppress air pollution in the pilot region. What are the impact mechanisms involved? We next examine the three potential mechanisms mentioned in the theoretical analysis. Since mediation effects models based on triple regressions may suffer from endogeneity bias as well as poor identification of the influence channels, we draw on the method of Du et al. (2023) to conduct a mechanism analysis by examining the impact of the core independent variables ($Treat \times Post$) on the mechanism variables.

First, in terms of technological innovation, patents are typically used as indicators of urban innovation (Tong et al. 2014). Drawing on existing studies (Guo and Zhong, 2022; Jiang and Xu, 2023; Qiu et al. 2023; Hu et al. 2023b; Xu and Hu, 2024), this paper characterizes urban innovation (*Innovation*) in terms of the logarithm of invention patent applications. Columns (1)–(2) in Table 8 detail the corresponding results; the estimated coefficients of $Treat \times Post$ are positive at the 5% and 10% levels, respectively, revealing that the EQT pilot promotes urban technological innovation. Urban technological innovation effectively suppresses air pollution and industrial waste gases (Liu et al. 2022; Yang et al. 2021; Wang et al. 2023a). Therefore, the EQT pilot can suppress pollution emissions by promoting urban technological innovation. Hypothesis 2 is therefore verified.

Second, in terms of industrial structure upgrading, drawing on Li and Su (2016) and Zhao et al. (2021b), we characterize industrial structure upgrading (*Structure*) as the ratio of tertiary industry value added to secondary industry value added in cities. The larger this indicator is, the more service-oriented the industrial structure is (Zhang et al. 2020). In Table 8, columns (3)–(4) detail the corresponding results; the estimated coefficients of $Treat \times Post$ are positive at the 10% and 5% levels, revealing that the EQT pilot facilitates the upgrading of the industrial structure. The high share of highly polluting industries keeps air pollution a continuous problem in some Chinese cities (Cheng and Kong, 2022). Driving the allocation of resources from traditional industrial sectors to cleaner service industries can effectively address the source pollution caused by the heavy industrial structure, which is conducive to urban air improvement (Li and Li, 2022). Therefore, the EQT pilot can inhibit urban air pollution by facilitating changes in the industrial structure. Hypothesis 3 is therefore verified.

Table 7 Regression results for changing cluster levels.

Variables	Soot1 (1)	Soot2 (2)
$Treat \times Post$	−0.320*** (−2.62)	−0.304*** (−2.72)
Cons	5.981*** (13.34)	10.236*** (21.57)
Obs	2272	2272
R ²	0.876	0.849
Control	Yes	Yes
City FE	Yes	Yes
Year FE	Yes	Yes

Note: (1) *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; (2) The values in parentheses are the t-statistics corresponding to cluster-robust standard errors at province $\times$ year.

Table 8 Results of the potential mechanism test.

Variables	Innovation (1)	Innovation (2)	Structure (3)	Structure (4)	Energy (5)	Energy (6)
$Treat \times Post$	0.110** (2.23)	0.101* (1.87)	0.054* (1.66)	0.017** (2.05)	0.022** (2.12)	0.014 (1.32)
Cons	6.663*** (1384.61)	5.277*** (13.48)	−0.006** (−2.17)	1.699*** (23.50)	0.140*** (134.14)	0.175*** (3.61)
Obs	2272	2272	2272	2272	2272	2272
R ²	0.951	0.952	0.866	0.989	0.797	0.803
Control	N	Y	N	Y	N	Y
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y

Table 9 Results of the geographic location heterogeneity test.				
Variables	Eastern Soot1 (1)	Noneastern Soot1 (2)	Eastern Soot2 (3)	Noneastern Soot2 (4)
<i>Treat × Post</i>	−0.049 (−0.39)	−0.506*** (−3.42)	−0.046 (−0.36)	−0.474*** (−3.40)
Cons	6.613*** (5.71)	6.261*** (11.72)	11.70*** (10.08)	10.21*** (18.45)
Obs	688	1584	688	1584
R ²	0.890	0.851	0.888	0.833
Control	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Empirical p-value	0.000***		0.000***	

Third, in terms of energy structure optimization, the proportion of electricity consumption in energy consumption can indirectly characterize the cleanliness of a city's energy structure, as data on renewable energy consumption at the city level are unavailable, and the current renewable energy sources are based mainly on electricity output. Drawing on Song et al. (2022), we convert a city's electricity consumption to the standard coal equivalent and calculate its share of the city's total energy consumption (*Energy*). A larger value indicates a cleaner energy mix in the city. Columns (5)–(6) in Table 8 detail the corresponding results; the coefficient of *Treat × Post* is positive, but the significance is not stable, indicating that the effect of the EQT pilot on the optimization of the energy structure is still weak. Potential reasons are as follows. For one thing, energy structure transformation constitutes a systematic overhaul that requires coordinated adjustments to power supply layouts, grid dispatching, storage facilities, and end-use consumption patterns. Such transformation is inherently time-consuming and cannot deliver rapid results in the short run. For another, the green premium dilemma remains unresolved. Renewable power generation is volatile, while supporting facilities for grid integration and storage lag behind. As subsidies are phased out, profit margins are squeezed, and interprovincial energy transmission and circulation mechanisms remain incomplete. Furthermore, the environmental externalities of fossil fuels have not been fully internalized into market prices, weakening the price transmission efficiency of quota trading. These barriers collectively slow the large-scale deployment of renewable energy, thereby dampening the improving effect of the EQT pilot on energy structure optimization. Hypothesis 4 is not supported.

Further discussion

In the previous section, the basic relationship between EQT pilot and pilot city air pollution was explored, and three influence mechanisms were examined. Next, we further shed light on the effect of the EQT pilot on air pollution control in terms of asymmetric and spatial spillover effects.

Heterogeneity analysis. In accordance with the heterogeneity of geographic location, resource endowment, and fiscal pressure, the heterogeneous pollution reduction effects of the EQT pilot are explored by grouping cities according to whether they are eastern cities, energy-oriented cities, or high-fiscal-pressure cities. The empirical *p*-value obtained by 1000 replicates (bootstrap) indicates that the estimated coefficients of *Treat × Post* in each subgroup are significantly different.

Geographical location heterogeneity. The EQT pilot may have heterogeneous pollution reduction effects depending on the geographical location of the city. In accordance with the regional classification criteria of the National Bureau of Statistics, the grouped regression results are detailed in Table 9. The EQT pilot has a small and insignificant effect on air pollution in eastern cities ($\alpha_1 = -0.049, p > 0.1$; $\alpha_1 = -0.046, p > 0.1$), whereas it has a significant dampening effect on air pollution in central-western cities ($\alpha_1 = -0.506, p < 0.01$; $\alpha_1 = -0.474, p < 0.01$). The relatively modest and statistically insignificant effect of the EQT pilot on air pollution in eastern cities may reflect a saturation effect rather than policy ineffectiveness. Eastern cities, characterized by more advanced industrial structures and cleaner production technologies (Yang et al. 2023), have already achieved relatively low emission levels before the policy intervention. As a result, the marginal abatement benefit of further emission reductions through quota trading is inherently lower than in less developed regions, where the scope for improvement is substantially greater. This saturation pattern is consistent with the logic of diminishing marginal returns in environmental policy, suggesting that the effectiveness of market-based instruments may vary systematically with regional development stages. The regression results are analogous to those of Shao et al. (2023), who concluded that China's carbon emissions trading suppressed sulfur dioxide emissions in central-western cities but not in eastern cities.

Resource endowment heterogeneity. The EQT pilot may have heterogeneous pollution reduction effects depending on whether the city is energy-oriented. According to the documents issued by the four pilot provinces, the EQT pilot covers high energy-consuming enterprises (specifically, the scope of energy trading in Zhejiang Province is new energy-intensive projects, while in Fujian, Henan, and Sichuan Provinces, all enterprises in specific high-energy-consuming industries that have reached a certain threshold of energy consumption are included in the pilot program). Energy-oriented cities tend to be clustered with high-energy-consuming firms and therefore have a relatively large number of implementation entities. In addition, energy-oriented cities tend to adopt a highly resource-dependent rough development mode, which leaves more room for pollution reduction. Therefore, based on the above two points, the EQT pilot may have a stronger pollution reduction effect on energy-oriented cities. Drawing on the approach of Shao et al. (2023), a city is energy-oriented if its resource dependence is higher than the sample mean. Resource dependence is defined as the share of mining employment in the secondary sector in the number of urban units employed. Table 10 shows the grouped regression results; the EQT pilot has a dampening effect on air pollution in both types of cities ($\alpha_1 = -0.794, p < 0.01$; $\alpha_1 = -0.237, p < 0.05$; $\alpha_1 = -0.741, p < 0.01$; $\alpha_1 = -0.228, p < 0.05$), but the effect on energy-oriented cities is greater and more significant. The subgroup regression results validate our hypothesis.'

Financial pressure heterogeneity. The EQT pilot may have heterogeneous pollution reduction effects depending on the degree of financial pressure cities face. The EQT system is a market incentive-based environmental regulation that forces enterprises to save energy and reduce emissions. For cities facing high fiscal pressure, implementation of the policy may be hindered. On the one hand, the EQT pilot converts enterprises' environmental resource use from being unlimited and uncompensated to being rarified and economized, which means that enterprises need to eliminate their crude production methods and increase their production costs. Moreover, the EQT in the pilot phase is bound to energy-intensive enterprises, which are often essential contributors to local fiscal revenues under China's current fiscal

Table 10 Results of the resource endowment heterogeneity test.

Variables	Energy-oriented <i>Soot1</i> (1)	Non-energy-oriented <i>Soot1</i> (2)	Energy-oriented <i>Soot2</i> (3)	Non-energy-oriented <i>Soot2</i> (4)
<i>Treat</i> × <i>Post</i>	−0.794*** (−3.12)	−0.237** (−2.07)	−0.741*** (−3.29)	−0.228** (−2.07)
Cons	7.365*** (7.39)	6.145*** (11.19)	11.48*** (10.90)	10.50*** (18.94)
Obs	584	1688	584	1688
R ²	0.874	0.875	0.862	0.852
Control	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Empirical p-value	0.000***		0.000***	

Table 11 Results of the fiscal pressure heterogeneity test.

Variables	High-pressure <i>Soot1</i> (1)	Non-high-pressure <i>Soot1</i> (2)	High-pressure <i>Soot2</i> (3)	Non-high-pressure <i>Soot2</i> (4)
<i>Treat</i> × <i>Post</i>	−0.261 (−1.31)	−0.358*** (−2.71)	−0.266 (−1.39)	−0.350*** (−2.80)
Cons	5.757*** (8.32)	6.030*** (8.80)	9.407*** (13.19)	10.82*** (16.44)
Obs	872	1400	872	1400
R ²	0.862	0.878	0.827	0.852
Control	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Empirical p-value	0.059*		0.079*	

system. Therefore, cities with tight financing restraints may relax the enforcement of regulations or even experience collusion between the government and enterprises. On the other hand, in the EQT pilot, the pilot areas should increase their fiscal and tax incentive policies to support enterprises in carrying out energy-saving transformations. For cities facing high fiscal pressure, supporting the energy savings of enterprises is a high-end demand. Therefore, fiscal and tax incentives are likely to be weaker, which is not conducive to driving enterprises to address pollution and impairing the policy effect of the EQT pilot. We express fiscal pressure as the ratio of city fiscal expenditure to fiscal revenue: a city is considered a high-fiscal-pressure city when its fiscal pressure exceeds the average of all cities; otherwise, it is considered a low-fiscal-pressure city. Table 11 details the grouped regression results. The EQT pilot has a small and nonsignificant effect on air pollution in cities with high fiscal pressure ($\alpha_1 = -0.261$, $p > 0.1$; $\alpha_1 = -0.266$, $p > 0.1$), while it significantly inhibits air pollution in cities with low fiscal pressure ($\alpha_1 = -0.358$, $p < 0.01$; $\alpha_1 = -0.350$, $p < 0.01$). The regression results confirm our hypothesis.

Analysis of spatial spillover effects. The above section identifies the effect of the EQT pilot on pilot cities' air pollution using the

Table 12 Global Moran's index.

year	<i>Soot1</i> Moran'I	<i>Soot2</i> Moran'I
2013	0.389*** (9.903)	0.291*** (7.418)
2014	0.384*** (9.765)	0.294*** (7.473)
2015	0.385*** (9.788)	0.276*** (7.026)
2016	0.375*** (9.507)	0.275*** (6.981)
2017	0.400*** (10.144)	0.257*** (6.532)
2018	0.376*** (9.524)	0.158*** (4.062)
2019	0.388*** (9.835)	0.156*** (4.016)
2020	0.357*** (9.060)	0.166*** (4.266)

Note: (1) *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; (2) z value in brackets.

common DID approach and does not consider the effect of the EQT pilot on air pollution in neighboring cities. On the one hand, the energy constraints in the pilot areas may force local high energy-consuming industries to move to the surrounding areas, i.e., leading to a pollution proximity transfer effect, thus exacerbating air pollution in the surrounding cities. On the other hand, the EQT pilot drives industrial upgrading, technological innovation, and energy adjustment in the pilot areas, which may have a demonstration effect (Chen et al. 2023c) and lead to the improvement of cleaner production in the surrounding cities. The net impact of the EQT pilot on surrounding cities' air pollution depends on the relative strengths of these two aspects. Therefore, we take spatial factors into account and use spatial econometric modeling to conduct an extended analysis.

Global spatial correlation test. The premise of spatial regression is that the variables are spatially correlated. Therefore, we select the binary adjacency matrix and apply the global Moran's I index to test the spatial correlation of air pollution (*Soot1* and *Soot2*). Table 12 shows the corresponding results; the Moran's I values of *Soot1* and *Soot2* in each city from 2013 to 2020 are greater than 0 and are significant at the 1% level, which strongly rejects the null hypothesis that there is no autocorrelation. This finding indicates that there are significant spatial correlations in both *Soot1* and *Soot2*; therefore, it is reasonable to include spatial effects in this study.

SDID regression results. Considering the spatial spillover effect of the improvement in the air environment induced by the EQT pilot, we introduce the spatial lag term and apply the spatial regression method in combination with the DID approach to construct the SDID model to evaluate the effect of the EQT on air pollution. The SDID model is shown in Eq. (3):

$$Pollution_{it} = \alpha_0 + \rho W Pollution_{it} + \alpha_1 Treat_i \times Post_t + \gamma Control_{it} + \mu_i + \tau_t + \varepsilon_{it} \quad (3)$$

where W represents the spatial weight matrix. If there is a common boundary between two cities, the element is set to 1; otherwise, it is 0. $W pollution$ is the spatial lag term of the dependent variables. Here, ρ refers to the coefficient of the spatial spillover effect, which indicates the impact of air pollution in

Table 13 Results for SDID.		
Variables	Soot1 (1)	Soot2 (2)
<i>Treat × Post</i>	−0.272*** (−2.62)	−0.270*** (−2.71)
Cons	0.177*** (5.26)	0.160*** (4.71)
$\rho(\rho)$	0.219*** (12.34)	0.206*** (12.24)
σ^2_{ϵ}	−0.270** (−2.51)	−0.268*** (−2.61)
City FE	−0.054** (−2.26)	−0.047** (−2.27)
LR_Direct	−0.325** (−2.51)	−0.316*** (−2.61)
LR_Indirect	Y	Y
LR_Total	Y	Y
Control	Y	Y
City FE	Y	Y
Year FE	Y	Y
Obs	2272	2272
R ²	0.169	0.323

Note: (1) *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; (2) z values are in parentheses under robust standard errors.

surrounding areas on local air pollution. Since the coefficient ρ in the spatial model does not fully represent the real elasticity, we also need to decompose the direct effect and indirect effect to determine the spatial spillover effect. In this case, the direct effect represents the average impact of the EQT pilot on air pollution in the local city, the indirect effect (spatial spillover effect) reflects the average impact of the EQT pilot on air pollution in other cities, and the total effect reflects the average impact of the EQT pilot on all cities. Table 13 reports the results of the SDID approach. The coefficients of $Wpollution$ are significantly positive ($\rho = 0.177$, $p < 0.01$; $\rho = 0.160$, $p < 0.01$), which initially suggests that there is a significant spatial spillover effect of intercity air pollution; i.e., the more polluted the city is, the more polluted the surrounding cities are. The coefficients of $Treat \times Post$ are still significantly negative ($\alpha_1 = -0.272$, $p < 0.01$; $\alpha_1 = -0.270$, $p < 0.01$), which further validates the robustness of the core findings. Moreover, the inhibitory effect of the EQT pilot on air pollution may be overestimated if spatial spillover effects are not considered. Furthermore, the direct and indirect effects of EQT are significantly negative (LR_Direct = -0.270 , $p < 0.05$; LR_Indirect = -0.054 , $p < 0.05$; LR_Direct = -0.268 , $p < 0.01$; LR_Indirect = -0.047 , $p < 0.05$), which indicates that the EQT pilot not only suppresses local air pollution but also contributes to reducing air pollution in surrounding cities. The positive spatial spillover effect of the EQT pilot implies that the demonstration effect of industrial upgrading, technological innovation, and energy restructuring exceeds the pollution transfer effect. Therefore, on the whole, the EQT pilot does not lead to an unfavorable situation of a zero-sum game in terms of pollution control but rather facilitates pollution control across the board. Similarly, previous studies have verified that environmental pilot policies lead to pollution reduction in surrounding cities; for example, Chen et al. (2023c) reported that China's LCCP suppressed PM_{2.5} not only in local cities but also in neighboring cities.

Summary

Main conclusions. Using data for 284 Chinese cities from 2013 to 2020 and the DID approach, our study examines the effect of the EQT pilot on urban air pollution. The main conclusions emerging

from the empirical tests are as follows. First, the EQT pilot effectively suppresses pilot cities' air pollution. These core findings are stable after undergoing multiple robustness tests. Second, improving technological innovation and optimizing the industrial structure are impact mechanisms through which the EQT pilot influences air pollution abatement, while the mechanism of energy structure optimization does not hold, which may be related to the green premium as well as to the immature nature of renewable energy technology and the market. In terms of urban heterogeneity, the pollution reduction effect of the EQT pilot varies markedly according to the geographic location, resource endowment, and financial pressure of cities. The EQT pilot is more effective at reducing pollution in central-western cities, energy-oriented cities, and cities with low fiscal pressure. Finally, the EQT pilot has spatial spillover effects and can help surrounding cities reduce air pollution.

Policy implications. First, the advancement of a nationwide EQT market should be accelerated. Competent authorities and pilot regions should promptly summarize the experiences gained from the pilot programs, overcome institutional and technical barriers, and expedite the expansion of the EQT system from pilot cities to the entire nation. The goal is to establish a unified, efficient, and transparent national EQT market to maximize its environmental benefits.

Second, differentiated and targeted policy promotion strategies should be implemented. For cities in developed eastern regions, stricter energy quotas and higher emission reduction targets should be set, leveraging their economic and technological advantages to play a leading role. For energy-oriented and noneastern cities, appropriate technical and financial support should be provided to help them foster alternative industries and enhance their technological innovation capabilities, enabling them to better adapt to and utilize the quota trading policy for a green transition. For cities under high fiscal pressure, accompanying fiscal transfer payments or green financial instruments can be designed to alleviate financial strain during the initial stages of policy implementation, preventing diminished governance effectiveness due to insufficient funding.

Third, the collaborative support system for technological innovation and industrial structure optimization should be strengthened. On the one hand, special funds can be established to incentivize enterprises to engage in the research, development, and application of green technologies, more closely linking technological upgrades with quota allocation and trading mechanisms. On the other hand, the market-driven pressure created by quota trading, coupled with industrial policies, can accelerate the phasing out of backward production capacity and guide capital toward high-tech, low-energy-consuming green industries. Furthermore, future policy adjustments should focus on enhancing direct incentives within the quota trading system for the consumption of clean energy (such as natural gas and renewable energy), gradually forming stable expectations for investing in and integrating renewable energy, and thereby mobilizing the enthusiasm of all stakeholders.

Fourth, regional joint prevention and control mechanisms should be established to internalize spatial spillover effects. When the national market is designed, quota trading and cooperation among cities within certain geographical ranges should be encouraged, and spillover effects should be transformed into collective regional emission reduction benefits. Additionally, the exchange of environmental technologies and the sharing of best practices between pilot and nonpilot cities, especially those that are geographically adjacent, should be promoted to amplify the positive spillovers of the policy.

Theoretical contributions. First, this study expands the research scope of EQT policy effects. By empirically investigating the air pollution mitigation effect of the EQT policy, the research perspective shifts from the traditional “energy–climate” nexus to an “energy–environment–health” framework. This study contributes to the theoretical system for evaluating energy and environmental policies and broadens the understanding of the effects of EQT. Second, this research contributes to the theoretical analysis of the pollution reduction mechanisms of EQT. Through testing the three mechanisms of “technological innovation,” “industrial structure optimization,” and “energy structure optimization,” this study clearly defines the internal logic through which EQT influences air pollution. It thereby provides an effective theoretical reference for subsequent research examining the environmental impact mechanisms of regulatory policies. Finally, this paper supplements the theoretical connotations of the spatial spillover effects of environmental policies. This confirms that the EQT pilot generates positive spatial spillover effects by effectively curbing air pollution in neighboring cities. These findings extend policy effect research from the single-city level to the regional coordination level. The conclusions offer theoretical support for cross-regional collaborative governance and for constructing environmental policy systems for urban agglomerations.

Limitations and future research. First, our work examines the pollution reduction effect of the EQT pilot using cities as samples, and we can further collect data on new and existing projects across pilot regions, along with samples of enterprises participating in EQT, to provide more solid micro evidence on the pollution reduction effect of EQT. Second, this paper explores only three potential mechanisms through which the EQT pilot achieves urban pollution reduction, and we intend to explore other possible paths to further open the black box of the impact mechanism. Third, this paper does not delve deeply into the policy design of EQT, and further research can be conducted to investigate the effects of the allocation of energy rights and the trading design on pollution reduction.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 18 June 2025; Accepted: 8 July 2026;

Published online: 20 July 2026

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Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 72373135).

Author contributions

SZ: Formal analysis, Writing—review, and Editing, CC: Methodology and Data curation. HZ: Resources and Software, FH: Conceptualization and Funding acquisition, YC: Investigation and Visualization. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Additional information

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